Unit-Roller485

I2C Control Protocol

Table of Contents

.....1
1、Communication Protocol Structure 3
1.1 Communication Protocol Parameters 3
2、Configuration Registers 3
2.1 Mode Switch(00H).....3
2.2 Mode Setting(01H).....3
2.3 Motor Over Range Protection(0AH).....4
2.4 Remove Protection(0BH).....4
2.5 Motor Status(0CH) 5
2.6 Motor Error(0DH).....5
2.7 Button Switch Mode(0EH).....5
2.8 Motor Jam Protection(0FH).....6
2.9 Device ID (10H).....6
2.10 RS485 Baud Rate(11H) 6
2.11 RGB LED Brightness(12H) 7
2.12 RGB LED Color(30H) 7
2.13 RGB LED Mode(33H) 7
3、Speed Loop Control Registers 8
3.1 Speed Setting(40H).....8
3.2 Speed Max Current Setting(50H) 8
3.3 Speed Readback(60H) 9
3.4 Speed PID Configuration(70H) 9
4、Position Loop Control Registers 10
4.1 Position Control(80H).....10
4.2 Position Max Current Setting(20H) 10
4.3 Position Readback(90H) 10
4.4 Position PID Configuration(A0H) 11
5、Current Loop Control Instruction Set 11
5.1 Current Control(B0H) 11
5.2 Current Readback(C0H).....12
6、Status Read Registers 12
6.1 Power Vin(34H) 12
6.2 Internal Temperature(38H) 13
6.3 Encoder Counter(3CH) 13
6.4 Save Flash(F0H) 13
6.5 Firmware Version (FEH).....14
6.6 I2C Address (FFH) 14

1、Communication Protocol Structure

1.1 Communication Protocol Parameters

The I2C communication interface is used.
Default Address is 0x64
Recommended communication speed: 200-400KHz.

2、Configuration Registers

2.1 Mode Switch(00H)

- Function: Motor enable switch
- Register Address:00H

Address	R/W	Length	Parameter
00	R/W	1 byte	Status

- Input Parameters:

  Status (1byte):

Parameter	Function	description
0x00	Motor Disable	Motor off
0x01	Motor Enable	Motor on

2.2 Mode Setting(01H)

- Function: Set motor operating mode.
- Register Address:00H

Address	R/W	Length	Parameter
01	R/W	1 byte	Mode

- Input Parameter:

  Mode (1byte):

Parameter	Function	Description

0x01	Speed Mode	Speed control loop: controls the motor to run at the target speed.
0x02	Position Mode	Position control loop: controls the motor to rotate to the specified position.
0x03	Current Mode	Current control loop: controls the motor to operate at the target working current.
0x04	Encoder Mode	Encoder mode: device acts as an input device to collect current encoder values.

2.3 Motor Over Range Protection(0AH)

- Function: Set motor rotation range protection. When enabled, if the encoder value is <-2100000000 or >2100000000, the motor stops and enters protection state.
- Register Address: 0AH

Address	R/W	Length	Parameter
0A	W	1 byte	Protection

- Input Parameters:

  Protection (1byte)：

  0： Disable rotation range protection

  1： Enable rotation range protection

2.4 Remove Protection(0BH)

- Function: Remove Jam protection. When the jam protection is triggered, send this command to unlock.
- Register Address: 0BH

Address	R/W	Length	Parameter
0B	W	1 byte	Status

- Input Parameters:

  Status (1 byte):

1: Remove Jam Protection

2.5 Motor Status(0CH)

- Function: Motor working status.
- Register Address: 0CH

Address	R/W	Length	Parameter
0C	R	1 byte	Status

- Input Parameter:

Status (1byte):

0: Standby

1: Running

2: Error

2.6 Motor Error(0DH)

- Function: Motor error status codes.
- Register Address: 0DH

Address	R/W	Length	Parameter
0D	R	1 byte	Status

- Input Parameter:

Status (1byte):

1: Overvoltage

2: Jam

4: Over Range

2.7 Button Switch Mode(0EH)

- Function: Enable button mode switching.
- Register Address: 0EH

Address	R/W	Length	Parameter
0E	R/W	1 byte	Status

- Input Parameter:

Status (1byte):

0: Disable button mode switching

1: Enable button mode switching (long press for 5 seconds to switch motor working mode)

2.8 Motor Jam Protection(0FH)

- Function: Enable or disable motor jam protection. When jam protection is triggered, the motor locks and stops rotating. Send the Remove Protection command to unlock.
- Register Address: 0FH

Address	R/W	Length	Parameter
0F	R/W	1 byte	Status

- Input Parameters：

Status (1byte)：

0: Disable jam protection

1: Enable jam protection

2.9 Device ID (10H)

- Function: Set device ID.
- Register Address: 10H

Address	R/W	Length	Parameter
10	R/W	1 byte	ID

- Input Parameters:

ID (1byte)：

0-255: New device ID

2.10 RS485 Baud Rate(11H)

- Function: Set RS485 communication baud rate.
- Register Address: 11H

Address	R/W	Length	Parameter
11	R/W	1 byte	Baud

- Input Parameters:

Baud(1byte):

0: 115200 bps

1: 19200 bps

2: 9600 bps

2.11 RGB LED Brightness(12H)

- Function: Control the brightness of the RGB LED.
- Register Address: 12H

Address	R/W	Length	Parameter
12	R/W	1 byte	Brightness

- Input Parameters:

Brightness (1byte):

0-100

2.12 RGB LED Color(30H)

- Function: Control the color of the RGB LED.
- Register Address: 30H

Address	R/W	Length	Parameter
30	R/W	3 bytes	Color

- Input Parameters:

Color(3byte):

Byte0: RGB-B value

Byte1: RGB-G value

Byte2: RGB-R value

2.13 RGB LED Mode(33H)

- Function: Control the RGB LED operating mode.
- Register Address: 33H

Address	R/W	Length	Parameter
33	R/W	1 byte	Mode

- Input Parameters:

Mode(1byte):

0: Default system state display

1: User-defined control

3、Speed Loop Control Registers

3.1 Speed Setting(40H)

- Function: Configure the target speed (RPM)
- Register Address: 40H

Address	R/W	Length	Parameter
40	R/W	4 bytes	Speed

- Parameters:

Speed (4byte):

Speed Setting = Speed Setting-byte0 + Speed Setting-byte1 * 256 + Speed Settingbyte2 * 65536 + Speed Setting-byte3 * 16777216

Actual Speed Setting = Speed Setting / 100

Range: -2100000000 ~ +2100000000

3.2 Speed Max Current Setting(50H)

- Function: Configure the maximum current limit for the target speed.
- Register Address: 50H

Address	R/W	Length	Parameter
50	R/W	4 bytes	Max Current

- Parameters:

Max Current (4byte):

Max Current = Max Current-byte0 + Max Current-byte1 * 256 + Max Current-byte2 * 65536 + Max Current-byte3 * 16777216

Actual Max Current = Max Current / 100

Range: -120000 ~ +120000

3.3 Speed Readback(60H)

- Function: Read the current motor speed value (RPM)
- Register Address: 60H

Address	R/W	Length	Parameter
60	R	4 bytes	Speed Readback

- Parameters:

Speed Readback (4byte):

Speed Readback = Speed Readback-byte0 + Speed Readback-byte1 * 256 + Speed Readback-byte2 * 65536 + Speed Readback-byte3 * 16777216

Actual Speed Readback = Speed Readback / 100

3.4 Speed PID Configuration(70H)

- Function: Configure PID parameters for the speed control loop.
- Register Address: 70H

Address	R/W	Length	Parameter
70	R/W	4 bytes	P
74	R/W	4 bytes	I
78	R/W	4 bytes	D

- Parameters:

P / I / D (4 byte):

PID setting value = PID-byte0 + PID-byte1 * 256 + PID-byte2 * 65536 + PID-byte3 * 16777216

P setting value = P * 10e5 = P * 100000

I setting value = I * 10e7 = I * 10000000

D setting value = D * 10e5 = D * 100000

4、Position Loop Control Registers

4.1 Position Control(80H)

- Function: Configure the target position.
- Register Address: 80H

Address	R/W	Length	Parameter
80	R/W	4 bytes	Position

- Parameters:

Position (4byte):

Position Setting = Position Setting-byte0 + Position Setting-byte1 * 256 + Position Setting-byte2 * 65536 + Position Setting-byte3 * 16777216

Actual Position Setting = Position Setting / 100

Range: -2100000000 ~ +2100000000

4.2 Position Max Current Setting(20H)

- Function: Configure the maximum current limit for the target position.
- Register Address: 20H

Address	R/W	Length	Parameter
20	R/W	4 bytes	Current

- Parameters:

Max Current (4byte):

Max Current = Max Current-byte0 + Max Current-byte1 * 256 + Max Current-byte2 * 65536 + Max Current-byte3 * 16777216

Actual Max Current = Max Current / 100

Range: -120000 ~ +120000

4.3 Position Readback(90H)

- Function: Read the current position value.
- Register Address: 90H

Address	R/W	Length	Parameter
90	R	4 bytes	Position Readback

- Parameters：

Position Readback (4byte)：

Position Readback = Position Readback-byte0 + Position Readback-byte1 * 256 + Position Readback-byte2 * 65536 + Position Readback-byte3 * 16777216

Actual Position Readback = Position Readback / 100

4.4 Position PID Configuration(A0H)

- Function: Configure PID parameters for the position control loop.
- Register Address: A0H

Address	R/W	Length	Parameter
A0	R/W	4 bytes	P
A4	R/W	4 bytes	I
A8	R/W	4 bytes	D

- Parameters:

P / I / D (4 byte)：

PID setting value = PID-byte0 + PID-byte1 * 256 + PID-byte2 * 65536 + PID-byte3 * 16777216

P setting value = P * 10e5 = P * 100000

I setting value = I * 10e7 = I * 10000000

D setting value = D * 10e5 = D * 100000

5、Current Loop Control Instruction Set

5.1 Current Control(B0H)

- Function: Configure the target operating current.
- Register Address: B0H

Address	R/W	Length	Parameter

B0	R/W	4 bytes	Current

- Parameters：

Current (4byte)：

Current Setting = Current Setting-byte0 + Current Setting-byte1 * 256 + Current Setting-byte2 * 65536 + Current Setting-byte3 * 16777216

Actual Current Setting = Current Setting / 100

Range: -120000 ~ +120000

5.2 Current Readback(C0H)

- Function: Read the current operating current.
- Register Address: C0H

Address	R/W	Length	Parameter
C0	R	4 bytes	Current

- Parameters:

Current (4byte)：

Current Readback = Current Readback-byte0 + Current Readback-byte1 * 256 + Current Readback-byte2 * 65536 + Current Readback-byte3 * 16777216

Actual Current Readback = Current Readback / 100

6、Status Read Registers

6.1 Power Vin(34H)

- Function: Read the current input voltage value of the motor (V). The read value is VIN*100. Refer to the formula below to calculate the actual input voltage.
- Register Address: 34H

Address	R/W	Length	Parameter
34	R/W	4 bytes	VIN

- Parameters:

VIN(4byte)：

```
VIN X100 = VIN X100-byte0 + VIN X100-byte1 * 256 + VIN
X100-byte2 * 65536 + VIN X100-byte3 * 16777216

Actual VIN = VIN X100 / 100
```

6.2 Internal Temperature(38H)

- Function: Internal reference temperature value register (°C). The temperature is for internal status reference only and may not be accurate.
- Register Address: 38H

Address	R/W	Length	Parameter
38	R/W	4 bytes	Temperature

- Parameters:

Temperature(4byte):

```
Temp = Temp-byte0 + Temp-byte1 * 256 + Temp-byte2 * 65536 +
Temp-byte3 * 16777216
```

6.3 Encoder Counter(3CH)

- Function: In encoder operating mode, read the encoder value.
- Register Address: 3CH

Address	R/W	Length	Parameter
3C	R/W	4 bytes	Encoder Counter

- Parameters:

Encoder Counter (4byte):

```
Encoder Counter = Encoder Counter-byte0 + Encoder
Counter-byte1 * 256 + Encoder Counter-byte2 * 65536 + Encoder
Counter-byte3 * 16777216
```

6.4 Save Flash(F0H)

- Function: Save parameters to flash.
- Register Address: F0H

Address	R/W	Length	Parameter
F0	W	1 byte	Save

- Parameters:

Save (1byte): 1

6.5 Firmware Version (FEH)

- Function: Firmware version register.
- Register Address: FEH

Address	R/W	Length	Parameter
FE	R	1 byte	Version

- Parameters:

Version(1byte):

1-127

6.6 I2C Address (FFH)

- Function: Change I2C address.
- Register Address: FFH

Address	R/W	Length	Parameter
FF	R/W	1 byte	Address

- Parameters:

Address(1byte):

1-127